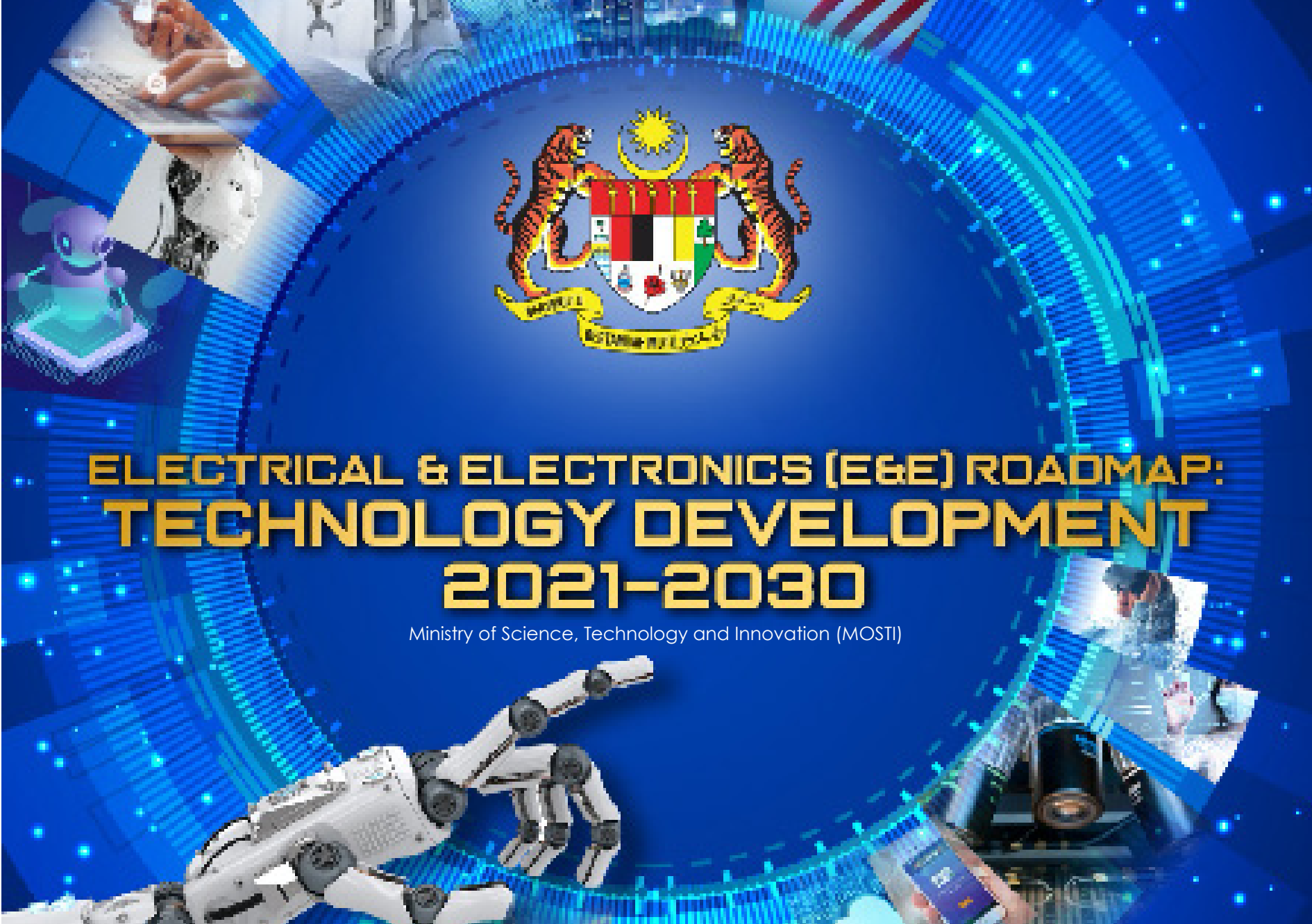


Ministry of Science, Technology and Innovation
(MOSTI)

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ELECTRICAL & ELECTRONICS (E&E) ROADMAP:
TECHNOLOGY DEVELOPMENT 2021-2030

Ministry of Science, Technology and Innovation (MOSTI)

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1 PREFACE

EXECUTIVE SUMMARY

As shown in Figure 1.1, the Electrical & Electronics (E&E) sector contributed to 6.3% of the country's Gross Domestic Product (GDP), equivalent to over RM89.2 billion in 2019. The industry gave more than RM373.12 billion in export value representing 37.5% of all export, and 87.6% of its trade surplus.

It also employs more than half a million people and topped the approved investments in the manufacturing sector, totaling RM25.66 billion or 31% of the approved investments. The data indicate the growth and the importance of the E&E industry to the national economy has fallen relative to other industries of the manufacturing sector (12.2% contribution for GDP and 58.8% export contribution in 2000). However, improvements were recorded between 2015-2020 where the export ratio has increased from 55% to 60%.

The government aims to increase the value-added of local industry by more than 40% to achieve the level of a developed country, based on scientific advances and high technology innovation. Based on Table 1.1, throughout these five years from 2015 to 2019, employment in E&E industries has shown a significant increase. To further strengthen and accelerate industry growth in this sector, the Electrical & Electronics (E&E) Roadmap: Technology Development 2021-2030 is providing a strategic direction and action plan for the industry and relevant ministries and agencies.

While Malaysia is relatively strong in the midstream semiconductor value chain of assembly, test, and packaging, most activities are still mainly driven by multinational companies with lower participation from local companies. Local participation in upstream development, for example in materials, fabrication and IC design as well as downstream development in components, modules and final products are insufficient despite significant efforts made in the past to foster industry development. Meanwhile, the global industry remains a highly competitive space with major players dominating by market share and fast-paced technology cycles create strong cyclicality that favours those larger players.

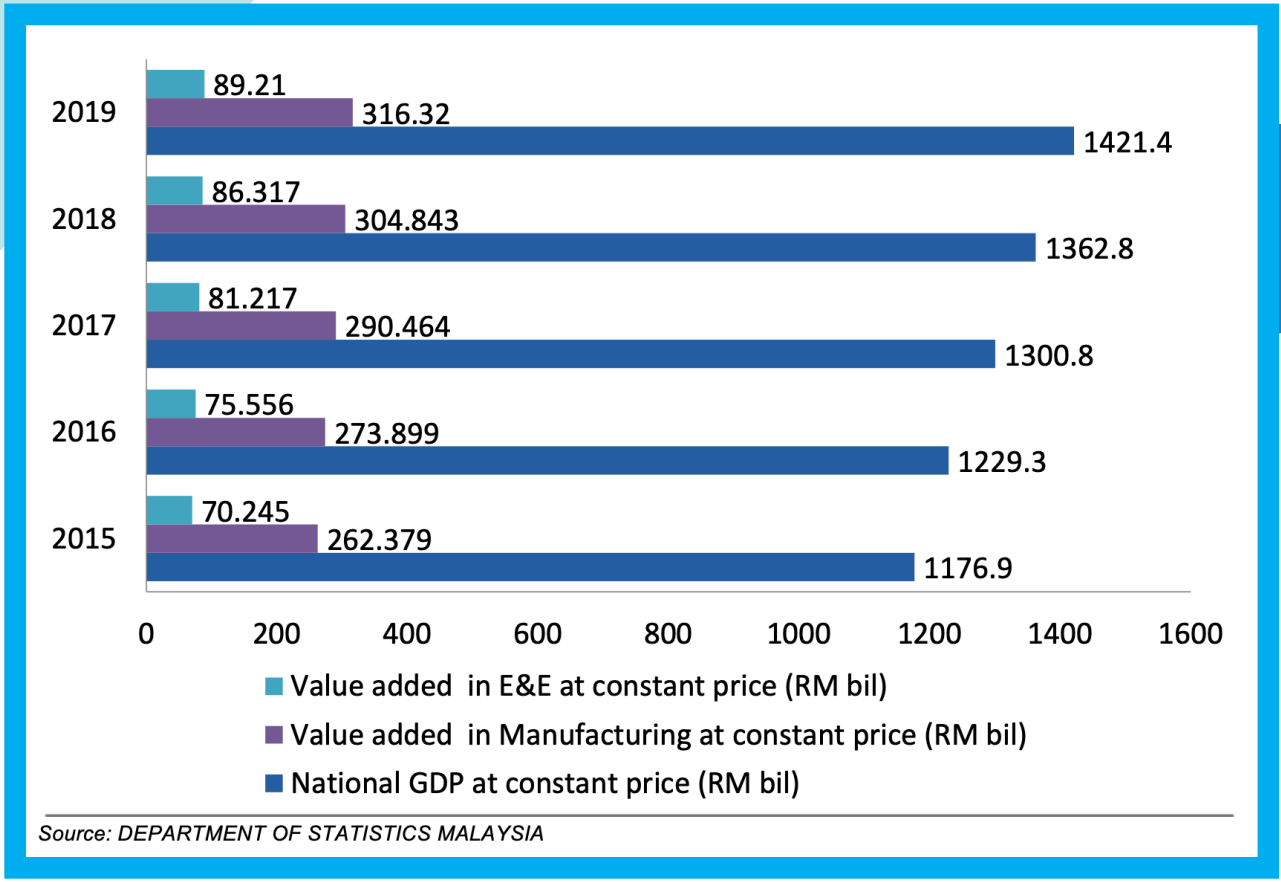


Figure 1.1 : Contribution of E&E value added to the GDP (2015-2019).

The Electrical & Electronics (E&E) Roadmap: Technology Development 2021-2030 will be a big part of the country's strategic direction to spur the growth of E&E sector, as shown in Figure 1.1. The formulation of this roadmap is for future development and growth of the country especially to develop strategies and action plans for 12th Malaysia Plan (12-MP) and E&E related endeavor through National E&E Council.

The target is to accelerate the growth of Electrical & Electronics (E&E) ecosystem and to strengthen the E&E sector as engine for economic growth beyond 2020

through technological development and establishment of local champion. The roadmap provides a deep-dive analysis on the future of E&E; one of the catalytic sectors outlined as high-technology industry in the SPV2030 and covers the analysis of current E&E industry situations and its status globally and in Malaysia, the key issues and challenges particularly related to technological development and its contribution to Malaysia's E&E ecosystem.

The roadmap also features some possible ways forward to sustain the E&E sector as a leading industry sector in Malaysia and propose its future direction for immediate action within the first five years (2021-2025) and following next five (5) years (2026-2030)

The Electrical & Electronics (E&E) Roadmap: Technology Development 2021-2030 consists of two important elements: strategic recommendations and strategic programmes as illustrated in Figure 1.3. In strategic

	2015	2016	2017	2018	2019
Employment total ('000 Persons)	14067.7	14163.7	14476.8	14776.0	15073.40
Employment in Manufacturing ('000 Persons)	2322.7	2390.6	2513.3	2499.9	2243.78
Employment in E&E ('000 Persons)	533.0	537.0	545.0	562.0	562.78

Source: DEPARTMENT OF STATISTICS MALAYSIA

Table 1.1 : Contribution of E&E to the total employment.

recommendations, there are five important cores which are investment and incentives, centralised one-stop centre, talent and skills, common platform, regulatory and innovation.

Strategic programmes are formed to strengthen the E&E sector and to achieve these strategic recommendations cores. Strengthen governance, ecosystem development, technology and innovation capacity development are the critical components under the strategic programmes.



Figure 1.2 : Electrical & Electronics (E&E) Roadmap: Technology Development formulation

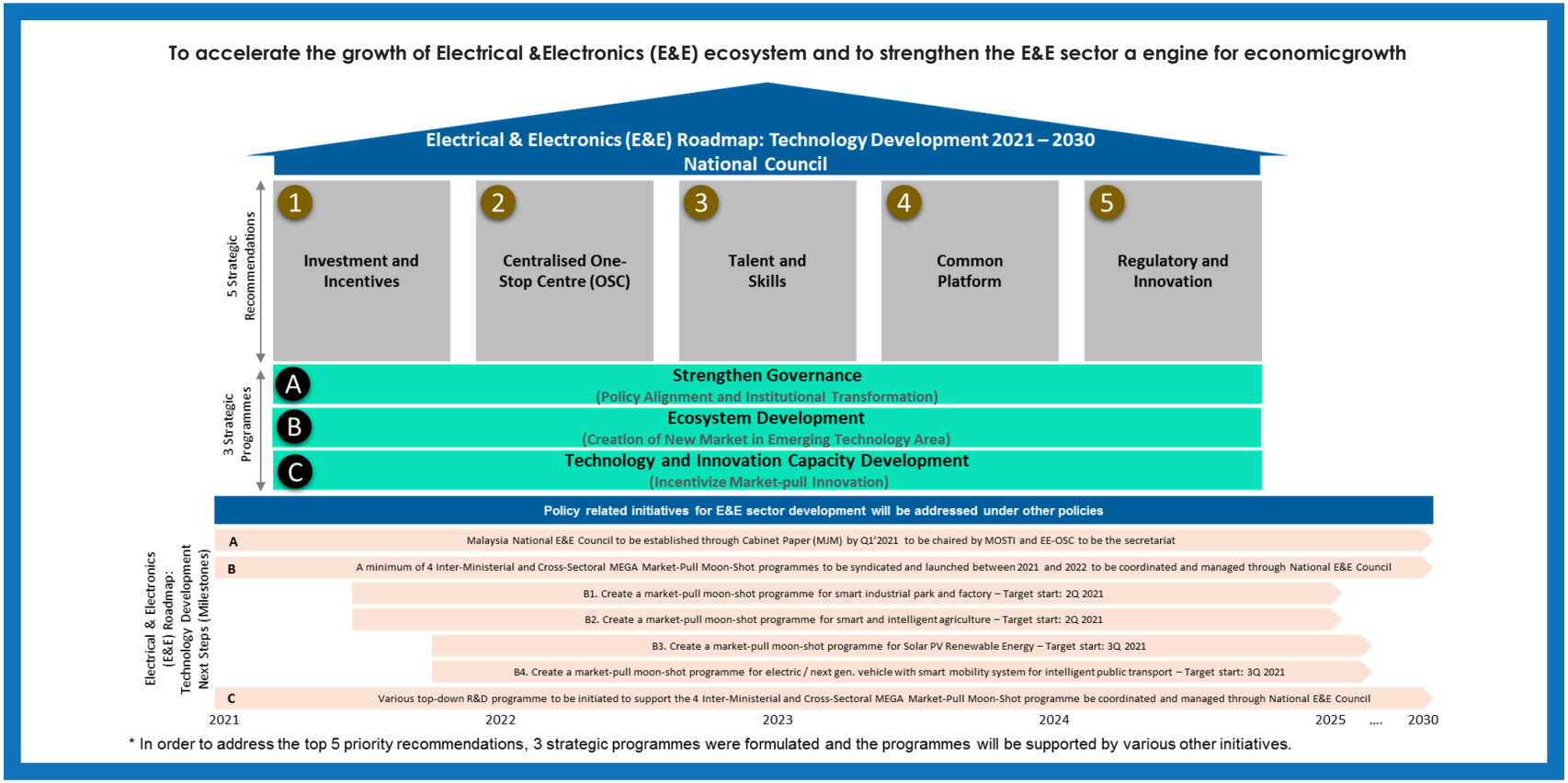


Figure 1.3 : Electrical & Electronics (E&E) Roadmap: Technology Development strategic formulation

2 INTRODUCTION TO MALAYSIA E&E SECTOR

2.1. Evolution of Malaysia E&E Industry

Figure 2.1 shows the evolution of the country's E&E industry. In general, the local E&E industry has shown great advances since its first establishment in the country. The core business of the industry in the early 1970s was the low-technology products that require a large amount of labour. It is foreseen that the Malaysian industry is capable of venturing into the high-technology product market with high value-added in the future.

2.2 Where is Malaysia's E&E Industry in the eyes of the world?

The semiconductor complexity and value by production are defined in three phases as described in Table 2.1. Design & development possess the highest technological complexity, medium capital intensity and a high share value of the final chip at 45%. Front-end manufacturing is a capital-intensive industry even though the level of technological complexity and share of the value of the final chip is the same as R&D. Among the three industries, back-end manufacturing has the lowest capital intensity and worth only a 10% share of the value of the final chip.

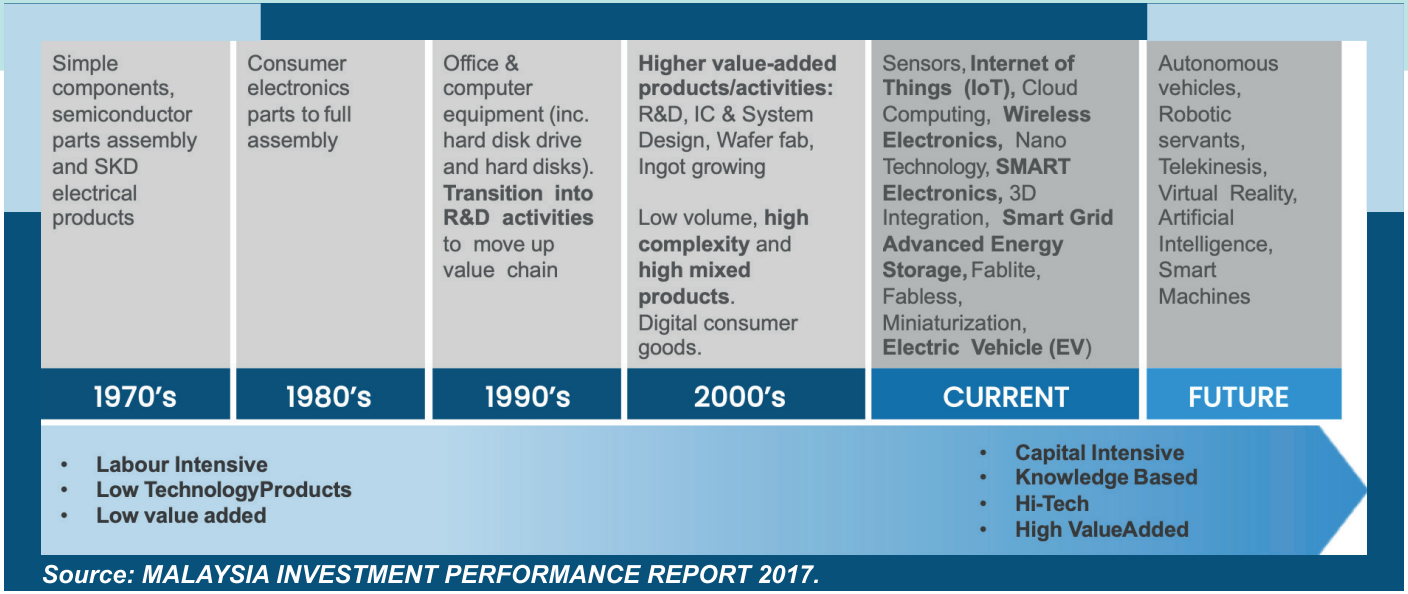


Figure 2.1 : Evolution of Malaysia E&E industry

	Design & Development	Front-End Manufacturing (Wafer Fabrication)	Back-End Manufacturing (Assembly, Test & Packaging)
Technological Complexity	High	High	Medium
Capital Intensity	Medium	High	Low
Share of Value of Final Chip	45%	45%	10%

Source: ANALYSIS BY BELFER CENTRE, HARVARD KENNEDY SCHOOL (JULY 2020).

Table 2.1 : Semiconductor complexity and value by production phase

In the semiconductor value chain, Malaysia is currently leading in the sector of back-end manufacturing involving assembly, testing, and packaging as seen in Table 2.2.

	Architecture & Design	Front-End Manufacturing (Wafer Fabrication)	Back-End Manufacturing (Assembly, Test & Packaging)	Equipment & Materials
US				
UK				
Netherlands				
Germany				
China				
South Korea				
Japan				
Taiwan				
Malaysia				
Philippines				

Capability

None

Minimal

Strong

Leading

Source: EEPN

Table 2.2 : Production capabilities by countries. E&E industry in Malaysia



Figure 2.2 : Local E&E sector in Malaysia

The local E&E sector which contributes about 6.3% to Malaysia's gross domestic product can be classified into four sub-sectors, namely electronic components, consumer electronics, industrial electronics and electrical products. Activities and products related to these four subsectors are listed in Figure 2.2.

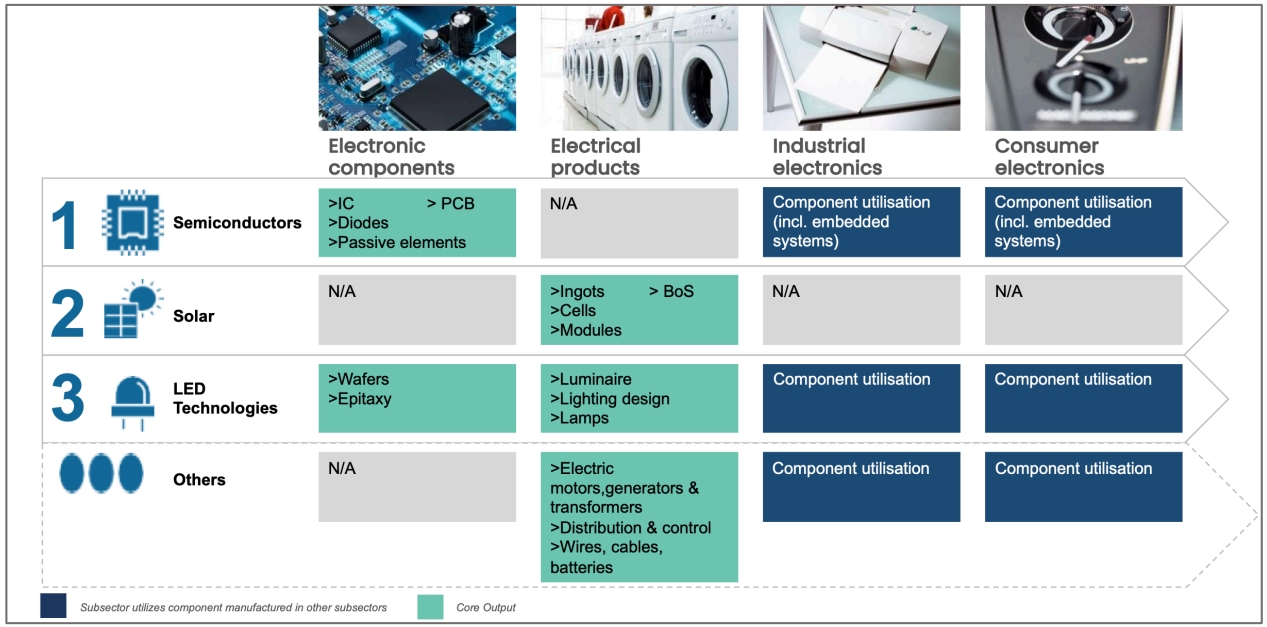
2.3 Key Ecosystem

Through its strategic initiatives, the government has created distinct electronics manufacturing clusters comprising of an "ecosystem" of semiconductor and components/equipment manufacturers. In other words, these subsectors are dependent on each other. Some of the subsectors utilise components that are manufactured by other subsectors. As illustrated in Figure 2.3, four major clusters have been identified i.e. Semiconductor, Solar, LED Technologies, and Others.



2.4 Malaysia E&E innovation driver

The R&D ecosystem in the country is relatively ineffective. Despite more than 40 years of industry history and several initiatives launched, limited mass scale success has been achieved. Malaysia has minor aggregate R&D expenditure roughly 1% as compared to peers in the E&E industry which is ~2% to 4%. Investment in R&D by corporates and industry-university partnerships are also limited. Malaysia also has modest commercialisation of R&D activities of work primarily focused on basic research. The majority of the industry talents are engaged in production/ process improvement activities than core R&D. Largely, Malaysia has inadequate utilisation of facilities of public sector R&D bodies. The illustration of the summary is shown in Figure 2.4.



Source: MIDA, ROLAND BERGER.

Figure 2.3 : Key ecosystems in the E&E industry in Malaysia.

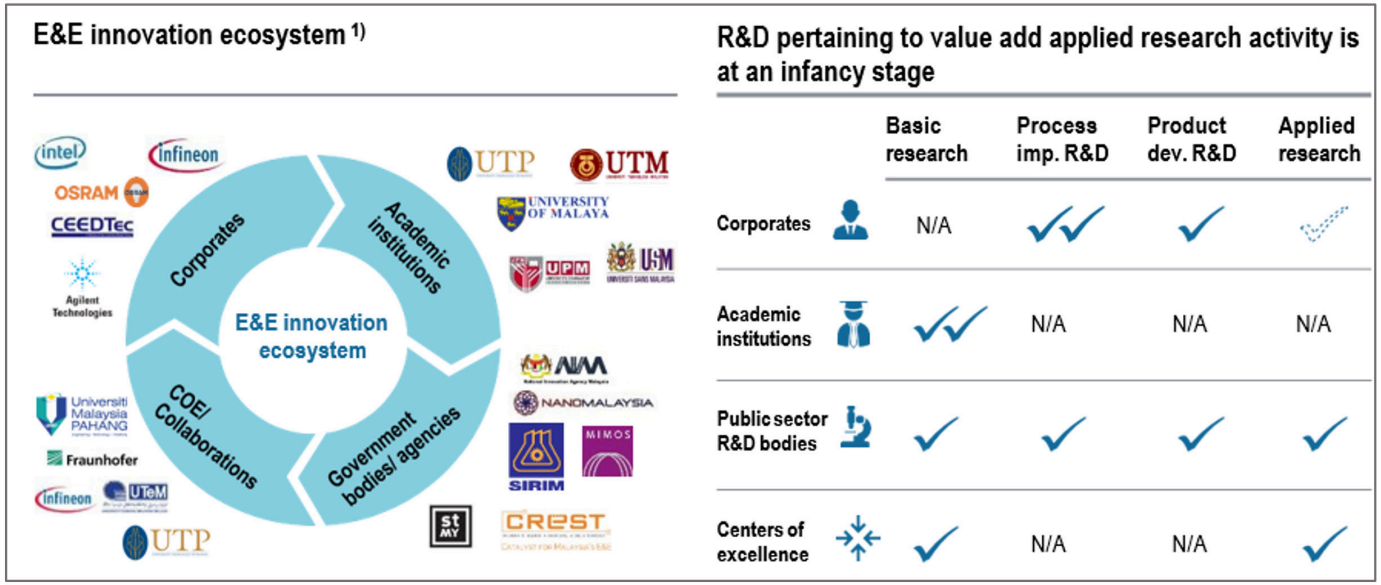


Figure 2.3 : Malaysia E&E innovation driver

2.5 The Need to Grow Malaysia E&E Sector

The growth of the E&E industry is very important to the national economy. The E&E sector contributed 6.3% of the country's GDP or equivalent to RM89.2 billion in 2019 The Value added of E&E from 2015-2019 in comparison with National GDP and also Value added of Manufacturing sector is as depicted in Figure 2.5.

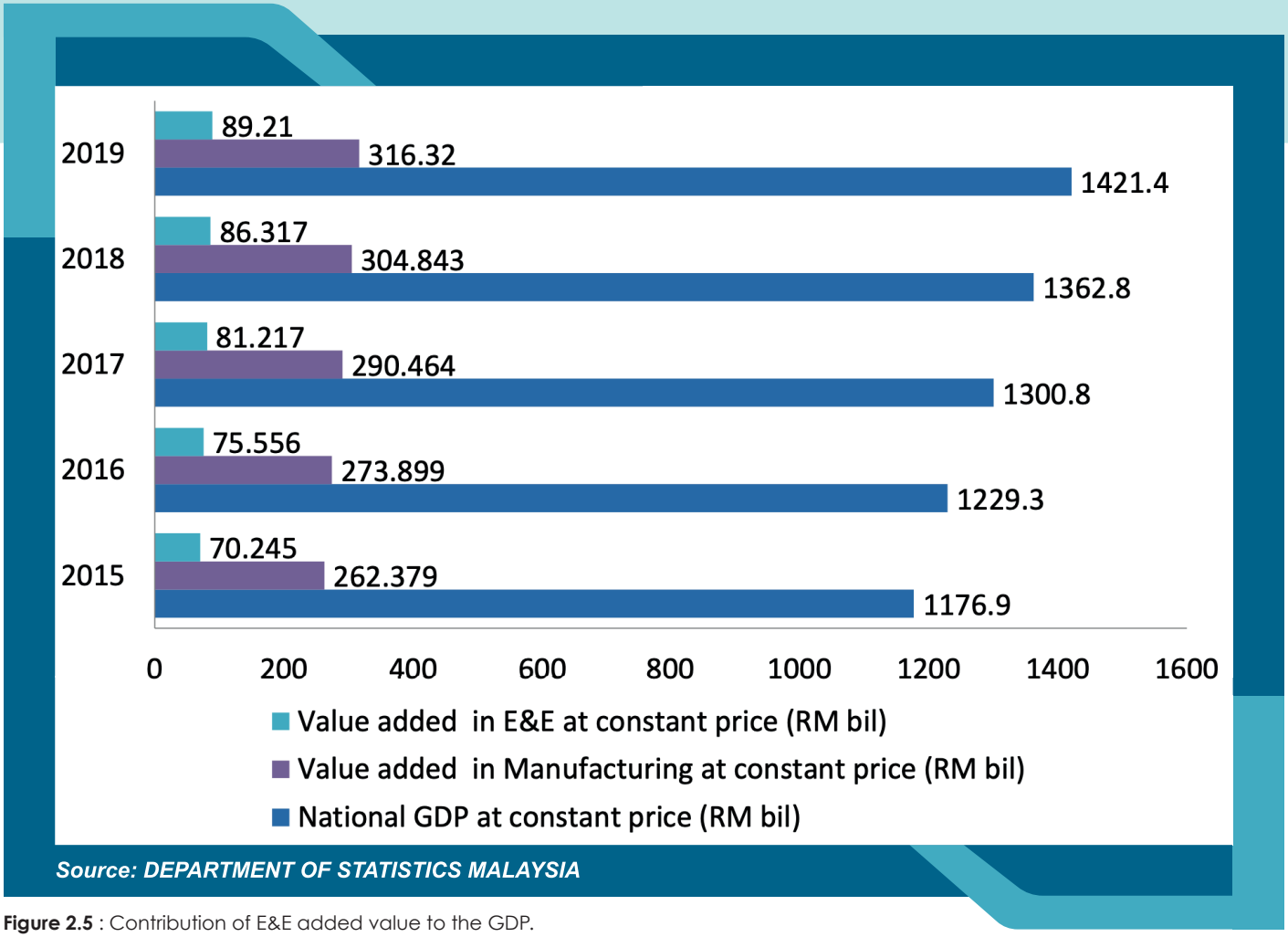


Figure 2.5 : Contribution of E&E added value to the GDP.

The E&E industry has contributed significantly to the country's overall export. The trend of the country's export is greatly affected by the local E&E growth. It contributed some RM373.12 billion in export value representing 37.5% of all export and 87.6% of its trade surplus in 2019 as show in Figure 2.6. On the other hand, the highest annual export occured in 2018 with total exports valued at RM 1 trillion – at which E&E total export was around 38.0%.

From 2015 until 2019, the E&E industry has contributed more than 85% of the country's total export surplus. 2019 was the year with the highest performance of the industry, in terms of financial trade, as the trade surplus increased from RM 118.92 billion in 2018 to approximately RM 127.66 billion in 2019. Unfortunately, there was a tremendous drop in the E&E total export surplus value in the three quarter of 2020.

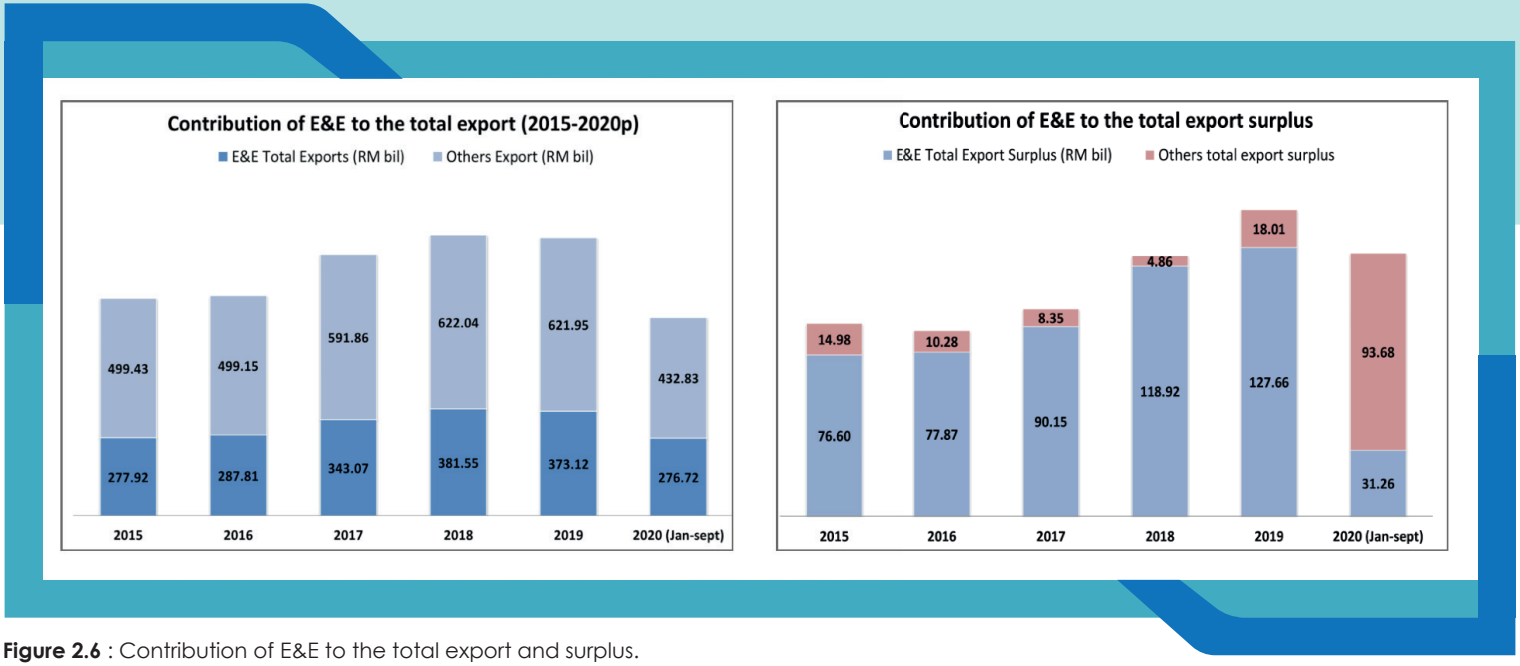


Figure 2.6 : Contribution of E&E to the total export and surplus.

For the investment, E&E sector topped the approved investments in the manufacturing sector in 2019, totaling some RM25.66 billion or 31% of approved investment.

These data indicates that the growth and the importance of E&E industry to the national economy has fallen relative to other industries of the manufacturing sector (12.2% contribution for GDP and 58.8% export contribution in 2000). However, some improvements were recorded between 2015-2020 (export ratio has improved from 55% to 60%).

The government aims to increase the value added of local industry by more than 40 per cent to achieve the level of a developed country, based on scientific advances and high tech innovation. To further strengthen and accelerate industry growth in this sector, the Electrical & Electronics (E&E) Roadmap: Technology Development 2021-2030 is providing a strategic direction and action plan for the industry and relevant ministries and agencies.

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Employment in E&E ('000 Persons)	533.0	537.0	545.0	562.0	562.78

Source: DEPARTMENT OF STATISTICS MALAYSIA

Table 2.3 : Contribution of E&E to total employment

E&E sector also employed 562,780 people in 2019 as seen in Table 2.3. Between 2015 and 2019, there are more than 14 million job opportunities created in the Malaysian market. Throughout this five-year period, the total number of employments increases steadily year by year which include the E&E sector. However, from 2018 there was a drop in the employment rate in the manufacturing industry alone. Nevertheless, the local E&E sector continues to grow as more employee were recruited, contributing to the increase of the overall employment.

3 TRANSLATING ASPIRATION TO PRACTISE

3.1 FIVE KEY STRATEGIC RECOMMENDATIONS

A comprehensive series of workshops and surveys have been conducted involving key stakeholders of the E&E sector in the country to provide input and feedback for the this roadmap since 2019. This involves various ministries, government agencies, E&E industry players including both multi-national companies (MNC) and Small and Medium Enterprises (SME) and also the universities. The outcome of the workshop and surveys are summarised in the five key strategic recommendations below. Five key strategic recommendations have been identified, as illustrated in Figure 3.1.

3.1.1 Recommendation one: Investment and Incentives

In Malaysia, the E&E Industry has an imbalance incentives ratio despite the sector's significant contribution to the Malaysian economy. Therefore, investment, incentives, funding, and grants are proposed to be allocated based on GDP contribution. In line with industry needs, the E&E contributed to 37.5% of export income, thus should enjoy a healthy corresponding allocation suggestions for investment incentives in Malaysia are compiled below:

i. Incentives should be weighted according to product/ process innovation & development, local manufacturing, and marketability

ii. Governmen R&D allocations should follow E&E GDP contribution preferably manage by Research Management Agency to address the mismatch of R&D between Universities, Research Institutes (RIs) and industry;

iii. Awareness by financial institutions on E&E SMEs be enhanced to reduce the difficulties in obtaining financing for SMEs in the E&E sector;

iv. Proper IP valuation process be implemented to overcome issues related to IP financing;

v. A platform for IP valuation for commercialisation, and royalty agreement be introduced to improve support between technopreneurs and Universities for technology ventures;



Figure 3.1 : Five key strategic recommendations of the Electrical & Electronics (E&E) Roadmap: Technology Development.

vi. Government focus on facilitating existing E&E industries to grow to the next level. Available incentives should also focus on established SMEs in new growth industries especially E&E;

vii. More attractive development and process-centric incentives for MNCs to create and develop local companies/vendors;

viii. More incentives on automation and robotics within the value chain;

ix. Effectiveness of the funds be reviewed and streamlined including the amount and its priority;

x. A one-stop centre to be set to speed up the process. Include a business growth-based funding mechanism.

xi. Incentivise local service providers to invest in new technologies to enhance their services.

xii. Incentivise new and existing pool of contract manufacturers in producing consumer appliances, medical and industrial electronic devices.

xiii. Monetisation for technology transfer of advanced IC packaging technologies from global technology providers, in expediting the acquisition of the necessary technologies.

xiv. Incentivise OSAT to invest in new process technologies and facilities to produce advanced IC packages.

xv. Retool existing government-owned semiconductor fabrication facilities to perform R&D on a Wafer–Level-Packaging (WLP) processes and product technologies.

xvi. Investment incentives given to SMEs to investing in Industry 4.0 technologies.

xvii. Incentives and support the marketing and distribution of new products domestically and in the export market.

viii. The government to provide incentive to GLICs undertaking CSR project related to advancing the E&E industry in Malaysia.

The incentives will be linked to specific outcomes, or specific E&E technology sub-sectors and related E&E technology new growth areas in emerging technologies.

In the long run, it is targeted for Malaysia's E&E industry players to become major players in the Global Electronics Value Chain (GEVC) and to have more "Malaysian-made" products and services.

3.1.2 Recommendation two: Centralised One-Stop Centre (OSC)

A centralised, one-stop centre to enhance support services and value chain development be established to provide shared services and Non-Recurring Engineering (NRE) for product development and R&D for the E&E sector in the country. Institutional support structure and ecosystem, coordination with the government and related stakeholders.

One-stop centre will help the government to understand priority issues across initiatives, ministries and agencies in charting effective programmes and providing necessary support to the centre. It is expected that the E&E One-Stop Centre that will govern all the National E&E initiatives mandated by the Malaysian cabinet and to be chaired by MOSTI as the lead ministries in the E&E Technology Development sector.

Activities suggested for the one-stop centre:

i. As shared facilities (test and certification for E&E products for a fast turnaround by consolidating existing publicly funded shared facilities to enhance effectiveness and utilisation by the industry;

ii. As a database of suppliers to promote local content in the value chain, it will be the platform for MNCs to know the local services and raw materials available by doing a stocktake of public companies and services. The database should have a feedback system to rate the suppliers/vendors;

iii. To complement Government efforts in developing product companies in both semiconductor and system product value chain by sharing Non-Recurring Engineerin (NRE) investments by Government / GLCs to support SMEs to reduce NRE cost for new products and to lead consortium approach to lower product cost for local SMEs (Penang Automation Cluster model). This complement can be accomplished by a one-stop D&D centre to enable SME E&E growth;

iv. Focus on key-value chain stages technologies and product components with higher chances of success (stages with the existing pool of competent companies). This key-value chain is to address the current issue of incomplete value chain due to "de-verticalisation" – companies began to focus on core competencies instead of vertically integrating production operations;

v. Promote Industry 4.0 technologies to help upgrade local industries using IoT and other technologies to grab data from a raw source. It shall help to establish industries standard communication protocol for customisable retrofit systems and solutions. This Industry 4.0 includes a standard communication protocol for a customised hardware solution;

vi. The centre should drive a partnership programme for larger SMEs to help develop smaller SMEs, mainly to accommodate the material that is not currently available locally. An example is creating a consortium of small SMEs, making them stronger to support larger SMEs, LLCs, and GLCs (e.g. SME Corp LED Consortium).

vii. Develop IC design and fabless IC clusters in Penang and the Klang Valley. Each group will consist of incubators with shared-facilities, including shared office space, shared IC design tools, shared IC test laboratories, and shared corporate services (HR/ admin/etc.).

vili. Establish IC design training centres and courses to train more IC designers by following the (previous) IC design training model implemented by SHRDC. IC test laboratories, and shared.

ix. Support for shared IP licenses to reduce the cost barrier of developing new ICs.

x. Establish shared facilities for product pre-compliance testing, such as CE, CB, and FCC

3.1.3 Recommendation three: Talent and Skills

The E&E industry in Malaysia is currently facing the challenges of the absence of a robust outfit for talent and skill development to produce more talents that are in line with industry needs.

There is an urgent need to create a skilled and diverse workforce with a high salary, both by upskilling the existing labour pool and attracting and developing future talent in the E&E sector.

Increase partnership between universities and industry players for the development of formal education relevant to industry and future workforce. engage industries for talent planning. Focus on the quality of students and

Suggestions are given below:

i. SIAP programme; a university enhancement and embedment programme for IC design facilitated by subject matter experts from MIMOS and CREST

ii. Specialised universities, e.g. Premier Digital Technology Institutions (PDTI) programme by MDEC – industries-certified Universities;

iii. Privately funded initiative – create an institution run by the industry (e.g. Forward School) for upskilling purposes;

iv. University in factory programme;

v. Local SMEs and universities collaboration.

vi. Upskill and reskill talent to be IR4.0 ready to ensure the workforce is relevant to future industry needs thus reduce the potential issues of unattractive remuneration for E&E workforce (diploma and below);

vii. An effort to be undertaken to share MNC/LLC Roadmap to prepare SMEs capacities (anchoring). One option is to identify ten anchor companies and look at their operational technology needs and ecosystem establishment. Refocus RIs and Universities to support this targeted co-funded technology development and localisation;

viii. Industry to assist with the creation of a suitable module for industrial training (refer Dyson Institute);

ix. Leverage on existing government agencies and universities competent centres, IPTA CEOs, e.g. Semiconductor institute (MIMOS); and Automotive Institute (MARii).

x. Incentivise companies that undertake their talent development;

xi. STEM needs to be addressed at secondary schools;

xii. Scholarships for engineering programmes;

xiii. The lower-income tax rate for STEM-related professionals; and

xiv. Propose a minimum of 6-month internship for both lecturers and students (subsidised programme) and IR4.0 courses in the university as part of Curriculum embedment to address industry gaps.

3.1.4 Recommendation four: Common Platform

Public-private partnerships (PPPs). The government aims to enhance entrepreneurial and ecosystem development through better engagement among Government agencies, PRIs, IHL and industries players, especially for SMEs. via a common platform possibly through a one-stop centre with focus on the following activities:

i. Help SMEs develop new products and expansion e.g. Incentives for MNCs to assist in developing SMEs and be the nucleus to enhance engagement/JV with MNCs/LLCs. This overcomes the short development cycle of E&E products and services demand companies for frequent reinvestment. It should be noted that lack of capital will hinder new product development.

ii. Intensify MNC's vendor development programmes to create a more extensive supplier base for the supply of locally produced E&E components and manufacturing services.

iii. Promote collaborations between MNCs and local companies to develop advanced manufacturing solutions for the GEVC.

iv. Foster the transfer of knowledge from MNCs to local companies through vendor development and other collaborative programmes.

v. Promote the outsourcing of production-related activities i.e. core or non-core by MNCs to external service providers.

vi. Create an awareness programme and database to strengthen investment ecosystem;

vii. Upgrade and strengthen the new and existing pool of contract manufacturers with R&D capabilities to develop new and innovative products.

viii. Support the development of standard/shared product chassis by consortiums or PRIs to enable OBMs to expedite the development of new products

ix. Develop local services for Fab manufacturing (MRS) for SMEs whereby MNCs and LLCs will assist

local SMEs by providing market opportunities and grow their products and technologies capabilities, e.g. collaborative development with MNCs and LLCs for products and equipment. Two possible approaches could be adopted; which are through intrapreneurs and subsidiaries.

x. Synergise linkages of Universities - Applied Research - Industry needs

xi. Develop a platform for IP valuation for commercialisation and royalty agreement

xii. Build an integrated manufacturing ecosystem through competent local suppliers equipped with Industry 4.0 manufacturing facilities.

xiii. Create a market for the remanufacturing and refurbishing automation equipment to make investments into Industry 4.0 automation more affordable and reduce financial institutions' risks in borrowing to SMEs to procure such equipment.

3.1.5 Recommendation five: Regulatory and Innovation

the government intervention in the form of policy guidance and regulation for local components and technologies content, particularly for adoption and market entry reference, in selected sectors within government control needs to be introduced.

The suggestions include:

i. Introduce innovation policies to address the systematic failures by focusing on networking and improving firm absorptive capacities;

ii. Establish a methodology to assess "innovation" for incentive purposes;

iii. To leverage on Free Trade Agreements (FTA) for government-linked companies (GLC) and public procurement (government projects) to create demand for local E&E products and service;

iv. Government to take an active role in creating a consortium and cluster through syndications and appoint MNCs as lead firms to drive the initiative or through a government-appointed agency or one-stop centre. These initiatives will reduce non-competitiveness of local SMEs in terms of pricing due to insufficient volume

v. GLCs procurement should consider the local SMEs products and outputs in which the government is to set a suitable threshold for local content. This GLCs procurement will reduce the challenges of GLCs, not supporting local SMEs. The government shall increase tax to companies who sourced supply solely from overseas;

vi. Government to intervene by reviewing the salary scale and encourage companies to provide better perks and benefits or introduce a competitive income tax structure for STEM-related employment, following Romanians example;

vii. Government to revamp the tax system for business growth incentive both monetary and service to ease business growth and enable a business to go for public listing;

viii. Revamp open approved permit (AP) system to favour local SMEs for E&E products

ix. Review and update investment policies and related legislation to enhance Malaysia's attractiveness to FDIs and to attract more DDIs into the E&E industry.

x. Ensure flexibilities in investment policies to allow for customisation of FDI/DDI incentives to meet the specific needs of investors in the E&E manufacturing sector and sub-sectors.

xi. Intensify FDI promotions to garner investments divesting out of less favourable economies

and those seeking to collaborate with Malaysian companies.

xii. Promote and incentivise the re-location of MNC's regional business service hubs and procurement offices to Malaysia.

xiii. Promote and incentivise global distributors of electronic components and electrical appliances to relocate their headquarters and regional operations in Malaysia.

xiv. Review and update investment policies and related legislation to ensure that FDI incentive packages also cover global/regional business services and distributors.

xv. Facilitate regulatory changes that could be required for new forms of services, e.g., custom regulations may need to be updated to operationalise logistic services using drones.

3.1.6 Summary

The E&E sector in Malaysia mainly faces five main challenges that are delaying it from further prospering which are:

1. despite being one of the key economic sectors, there is an insufficient factor of a dedicated industry centre to orchestrate further development and growth of the industry particularly in providing shared services and common tools for research and development;

2. inadequate robust outfit for talent and skills development to produce more talents that are in-line with industry needs and intensify industry participation and cross-fertilisation between academia and industry;

3. inavailability of a common platform for entrepreneurial and ecosystem development in providing easy access to various information such as potential partners, vendors and suppliers; government incentives and grants information; and venue to share knowledge, technologies and intellectual properties (IP);

4. ineffective government intervention in the form of policy guidance and regulation for local components and technologies content particularly for adoption and market entry reference in selected sectors within government control; and

5. imbalance incentives ratio despite the sector's significant contribution to the Malaysian economy relative to its size.

3.2 THREE KEY PROGRAMMES

3.2.1 Strengthen Governance

Electrical & Electronics (E&E) Roadmap: Technology Development 2021-2030 consists of two main components; strategic recommendations and strategic programmes as depicted in Figure 3.2. Three strategic programmes supported by various other initiatives were formulated to address the top five priority recommendations. Strengthen gaovernance, ecosystem development and technology and innovation capacity development are the three essential elements in strategic programmes.

Among the fundamental issues identified in formulating the Electrical & Electronics (E&E) Roadmap: Technology Development 2021-2030 is the absence of a central institutional setup to orchestrate a converging effort by various government departments and agencies in strengthening the sector.

It is expected that a one-stop-centre (Figure 3.3) to be established assisting the coordination of various inter-departmental and inter-agencies engagement and alleviate the concern of potential governmental related bureaucracy. It is also expected that it will resolve current need gaps such as in technology development and product innovation programme by providing easy access for shared services facilities, tooling and resources to lower industry R&D investment. Moreover, it will support in providing talent and relevant industry skills and expertise development programme by facilitating cross-fertilisation between academia and industry and intensifying industry participation.

The OSC is expected to produce a stream of market-pull deep-technologies such as integrated circuit (IC) design, advanced devices, semiconductor front-end technology modules and systems and development of materials and advanced processes. It will also support entrepreneurial and ecosystem development programme by providing knowledge, technology, and

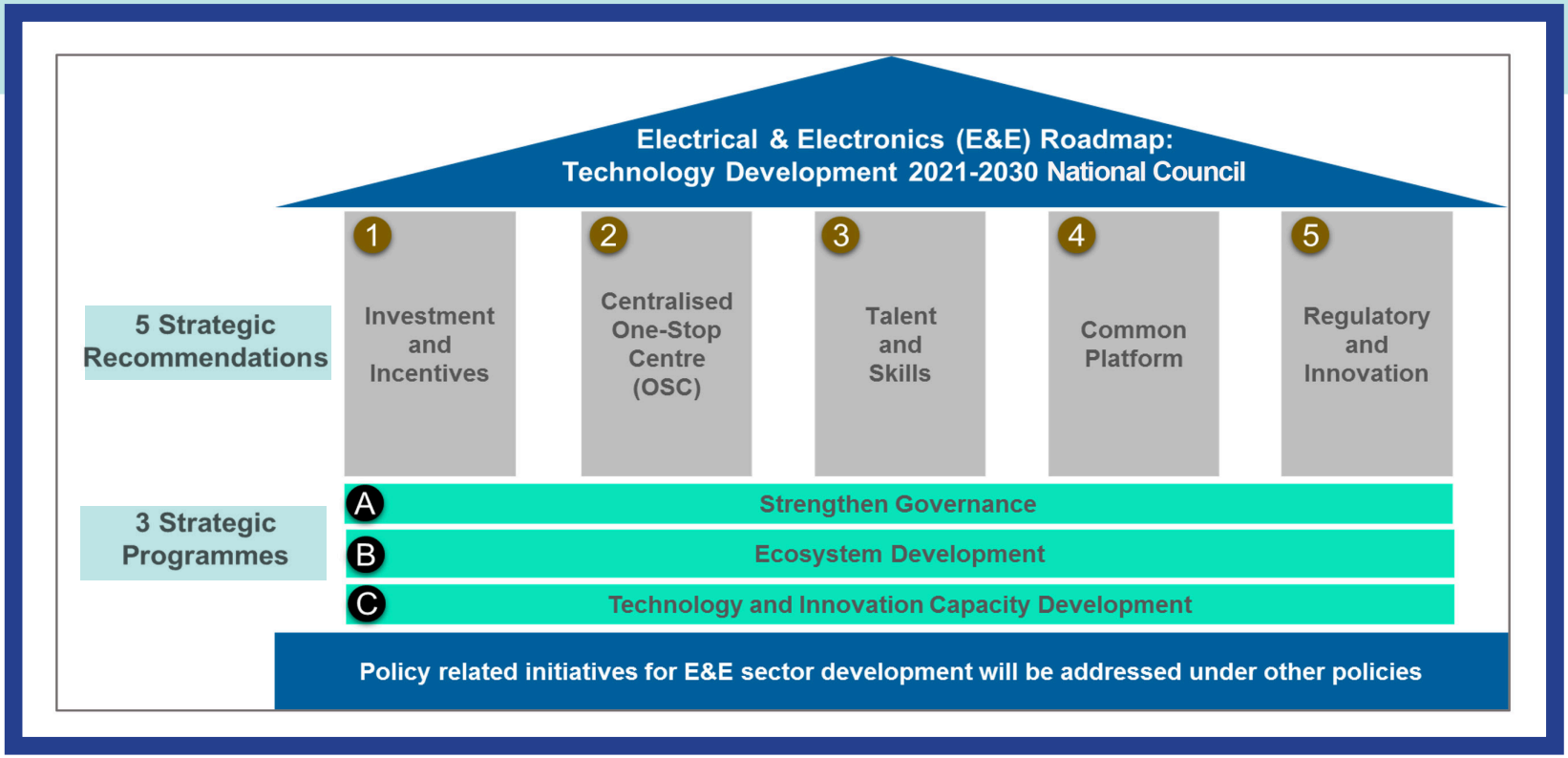


Figure 3.2 : Electrical & Electronics (E&E) Roadmap: Technology Development

Figure 4.2: Electrical & Electronics (E&E) Roadmap: Technology Development

2.2.2 EE-OSC Roles & Functions

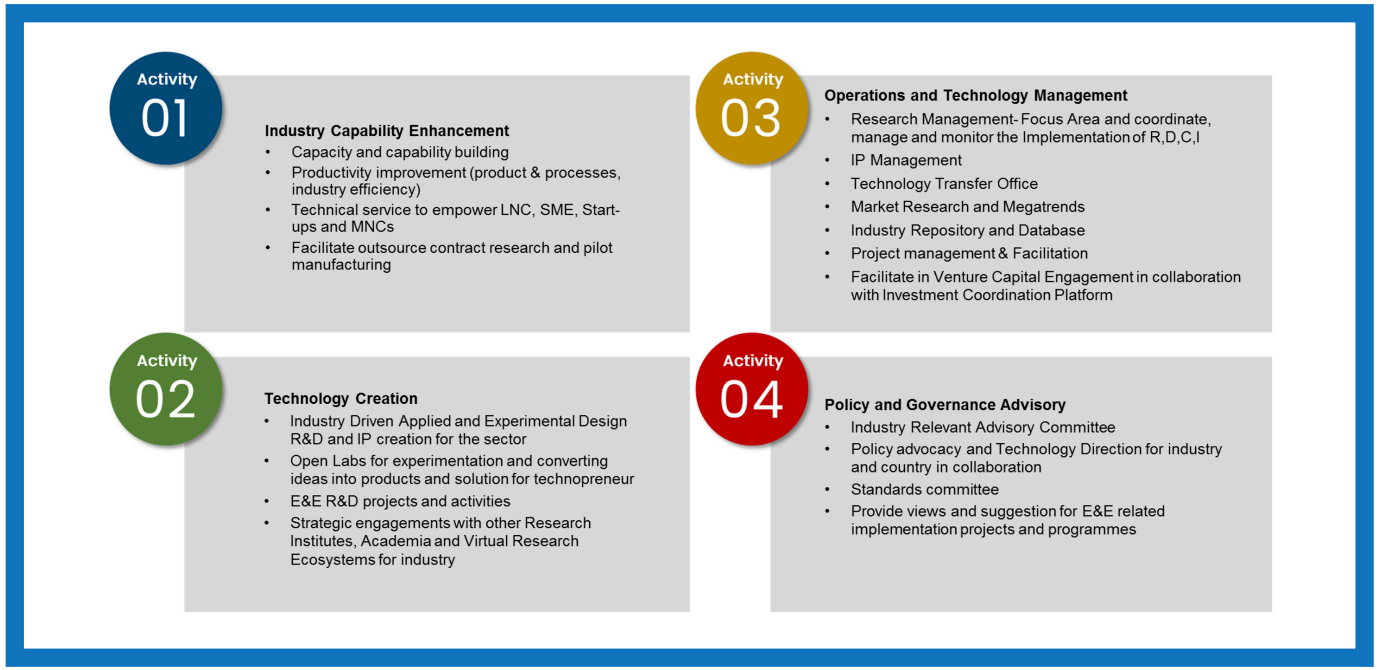


Figure 3.3 : EE-OSC activities

product incubation platform and outfit for partnership, value chain, and supply chain collaboration.

The centre is also expected to streamline and improve government support to industry through various technical advisory and advocacy activities such as in assisting multiple ministries and agencies in carrying out due diligence for incentives, funding and grants, industry improvement programme and syndication as well as to coordinate and monitor the implementation of Electrical & Electronics (E&E) Roadmap: Technology Development 2021-2030.

This initiative will facilitate local industry engagement with government through single window access and enable more local industry players to venture into related businesses and increase their capacity, competency and competitiveness by having quick access to various talent and technology developmental programmes. It is projected that a total of 500 new and emerging companies would be able to access the facilities and 5,000 new industry-relevant talents will be produced through a smart partnership with the industry.

3.2.3 Ecosystem Development.

A strategic inter-ministerial and cross-sectoral task force is needed to be formed to spur the local participation in new and emerging technologies; this task force will drive more vital local content requirement in the relevant demand sectors. Government efforts in trade facilitation such as countertrade, G2G engagement and global strategic partnerships can be used as a tool kit to help these companies become active contributors in the global value chain. Also promoting strategic foreign partnerships through encouraging equity participation where complementary capabilities may exist. These initiatives are expected to create monetisation effect of 1.0% from potential government projects equivalent to MYR 4.5 billion and industry spillover effect of over MYR 22.5 billion.

It is proposed for the E&E industry player to venture their technologies and products into other relevant vertical market and industry sectors which require E&E technologies and products components to improve their products performance and applications.

The market-driven developments provide a vast opportunity to enhance technology involvement through a market-pull based innovation model

3.2.4 Technology and Innovation Capacity Development

Future upstream and downstream value chain capabilities will need to be granularly identified, from comparisons to current capabilities. Critically, there is also a need to strike a balance between the aspired technology development agenda, with overall pace of industry development and impact to the economy. Therefore, detailed planning and pacing of local content targets across individual technology stacks are needed to achieve this balance.

In order to fully support guided capitalism initiative as above, more R&D allocation and funding is required to enable end-to-end R&D activities to be carried out and to ensure those R&D output would be able to go through extensive experimental development and testing programmes to meet the technology and products requirement standards. The guided capitalism initiative will eventually acculturate R&D mindset toward market-pull innovation whereby both upstream and downstream value chain capabilities will be identified and developed through syndication and participation of strategic sectoral and industry guides based on the sector emerging technology needs.

Another incentive is the two-dimensional inter-ministerial and cross-sectoral industry-driven technologies and product development efforts. This incentive ensures the successful execution of new technology and product commercialisation towards its sustainability for real industrialisation and market growth. This initiative aims to achieve 250 new technologies and product development initiatives to be eventually commercialised to the market.

Presently many government-funded R&D programmes either in Public Research Institutions or universities are geared toward advanced technology-push that may have resulted in imbalance with industry requirement resulted in constraint to comply with market need which entail complete product ecosystem, value chain and supply chain support. Additionally, most R&D grant and incentives are capital expense (CAPEX) based, hence the need to restructure R&D activities towards more operating expense (OPEX) based grants and incentives similar to other developed countries, as this would encourage market-pull activities in research and development.

This innovation initiative of E&E related cross-cutting technologies and products is also in line with other policies and roadmaps such as National 4IR Policy, Digital Economy Blueprint, National Science and Technology Policy (10-10 Science, Technology and Innovation for Economic Development Framework), Industry4WRD, Robotics, Artificial Intelligence, Blockchain, Advanced Materials, National Automotive Policy, and Green Technology Master Plan agenda.

3.3 TECHNOLOGY PRIORITY AREAS

3.3.1 The Malaysia E&E Key Pillars

Malaysia's E&E industry is supported by five focus areas, i.e., semiconductor, optoelectronics & LED, solar photovoltaic, system electronics, and manufacturing-related services. Every product/technology related to the E&E industry is grouped under its respective area, as shown in Fugure 3.4.

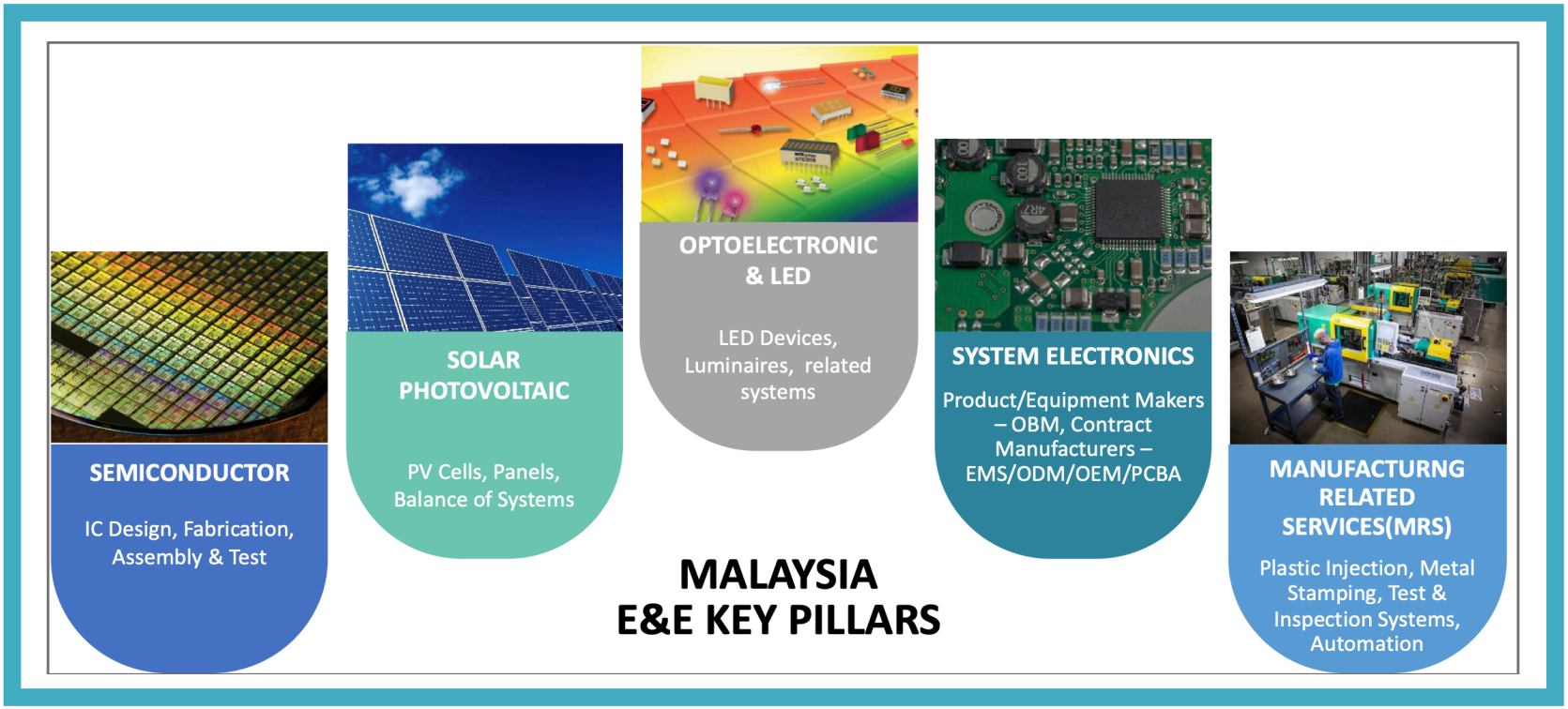


Figure 3.4 : Malaysia E&E key pillars.

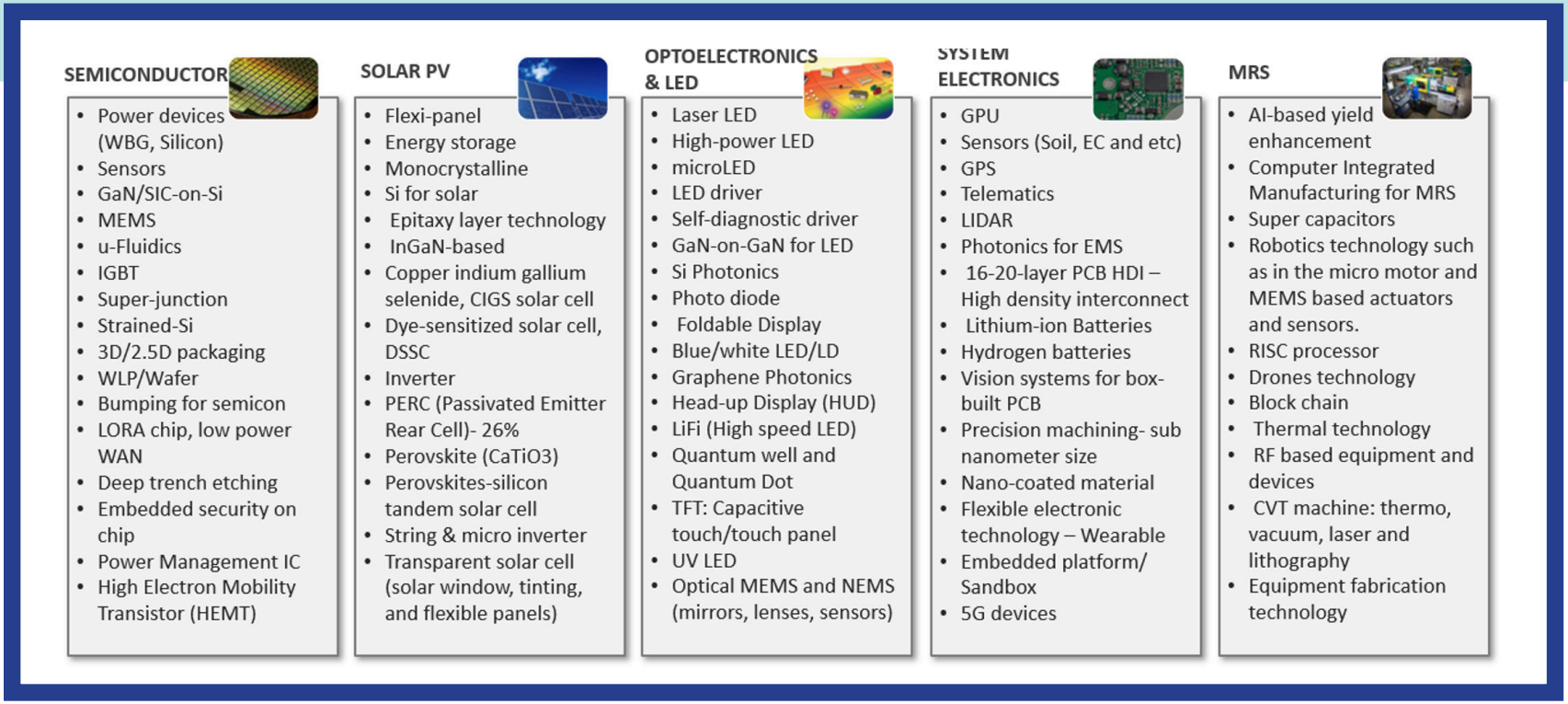


Figure 3.5 : Key Technologies Priority findings.

4. ROAD TO ACTION

SUMMARY

The Electrical & Electronics (E&E) :Technology Development 2021-2030 is developed to complement the country's E&E sector in order to identify the strategic core and main programs to increase 40% added value of the local industry based on science technology and innovation progression.

A proposal to strengthen the governance of E&E technology development through a centralized E&E sector governance structure sector that has the role of coordinating programs and initiatives related to E&E technology development.

Start-up program is proposed to support the development of the local ecosystem to create a market pull through the coordination of Government policies, strategies and investments that will enable more companies to participate in the implementation of projects in the identified areas such as renewable energy, smart manufacturing, smart agriculture and smart mobility systems for public transport.

Technology development and innovation capabilities to commercialize E&E technology and R&D will facilitate the E&E technology solutions of start-up companies, entrepreneurs, innovators and research institutions to penetrate the local market.



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